

National Academy of Science and Technology

(Affiliated to Pokhara University)

Dhangadhi, Kailali

Pre-University Examination

Level: Bachelor

Semester : VII_Fall

Year : 2025

Program: B.E. Computer

F.M. : 100

Course: Data Science and Analytics

P. M. : 45

Time : 3 Hrs.

Candidates are required to give their answer in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all questions.

1. a) Explain the Knowledge Discovery from Database (KDD) process in detail. How does it relate to the broader Data Science Pipeline? (7)
b) Distinguish between structured and unstructured data with appropriate examples. Why is data integration a challenging step in preprocessing data from multiple sources? (8)
2. a) Compare Numeric and Categorical variables. Explain how the Pearson correlation coefficient is used to measure the strength of a relationship between two numeric variables. (7)
b) Define a Power Law distribution and explain why it is referred to as a "heavy-tailed" distribution. How does it differ from a Normal distribution in real-world data? (8)
3. a) Describe the Multivariate Linear Regression model. Show the matrix formulation for the model and explain how the Ordinary Least Squares (OLS) estimation is used to find the coefficients. (8)
b) A researcher uses two predictors to model a target variable. Explain the significance of R-squared and Adjusted R-squared. Why is Adjusted R-squared preferred when comparing models with different numbers of predictors? (7)
4. a) Derive the Nadaraya-Watson kernel regression estimator. Discuss the critical role of the bandwidth parameter in determining the performance of this non-parametric model. (8)

OR

You are given the following dataset

x	0.5	1.0	1.5	2.0
y	-2.3	0.5	9.3	20.8

Estimate the value of y at $x = 1.3$ using the Nadaraya-Watson regression estimator with a bandwidth $h = 0.2$. (8)

b) Explain Principal Component Analysis (PCA) as a tool for identifying latent variables. Briefly explain the steps involved in it. (7)

5. a) Describe the CART (Classification and Regression Tree) algorithm. Explain the difference between Information Gain and the Gini Index as heuristics for attribute selection during tree induction. (7)

b) Illustrate the steps of the k-means clustering algorithm. How do we determine if the algorithm has reached convergence, and what are the limitations of using Euclidean distance for mixed-type data? (8)

OR

Apply the K-Means algorithm with $K=2$ on the following dataset: (160, 55), (175, 65), (172, 60), (180, 70), (178, 68), (165, 58). Use the first two data points (160, 55) and (175, 65) as the initial centroids. Perform the clustering for two iterations and show the cluster membership after each step. (8)

6. a) Discuss Logistic Regression and how it models the probability of a binary outcome. Why is it often preferred over linear regression for classification tasks like species habitat prediction? (7)

b) Define Autocorrelation and Stationarity in the context of time-series analysis. List and explain the four components of a time series: trend, seasonal, cyclical, and irregular variations. (8)

OR

Compare ARMA and ARIMA models. Explain why the "Integrated" (I) component is necessary when dealing with non-stationary data. (8)

7) Write short notes on (Any two): ($2 \times 5 = 10$)

a) Granger Causality

b) Statistical Significance and p-value

c) Akaike (AIC) and Bayesian (BIC) Information Criteria